MINDMAP-AI YOUR PERSONAL LEARNING NAVIGATOR

Project Description

- MindMap AI – Your Personal Learning Navigator is an AI-powered educational web application designed to help students organize, understand, and revise their study materials in an intelligent and personalized way. The system allows students to upload or enter learning materials such as notes, documents, and subject content, which are analyzed using Artificial Intelligence.

- The platform transforms complex and lengthy study materials into interactive mind maps, concise summaries, flashcards, quizzes, and personalized study plans. The mind-map feature organizes the main topics, subtopics, and related concepts into a visual structure, helping students understand the relationships between different concepts and improve their learning and revision process.

Scenario-Based Case Study

- **Consider a student preparing for multiple semester examinations while also learning new technical subjects. The student has study materials stored as PDFs, class notes, and other digital documents. Due to the large volume of content, the student finds it difficult to understand the complete syllabus and identify the relationships between important concepts.**

- The student wants to revise the subjects efficiently but spends considerable time manually creating notes, mind maps, flashcards, and practice questions. There is also no single platform to organize the learning materials, revision resources, study plans, and progress.

Problem

- Difficulty understanding lengthy study materials within limited time.
- Manual preparation of notes consumes a significant amount of study time.
- Creating flashcards and revision questions is repetitive and labor-intensive.

- Lack of personalized study plans based on available time and examination schedules.
- No centralized platform to securely store AI-generated learning resources.
- Difficulty assessing knowledge through self-generated quizzes.
- Switching between multiple applications for note-taking, quizzes, and scheduling reduces productivity.

Solution

The **AI StudyBuddy** backend provides a centralized, AI-powered learning platform that automates several academic tasks through intelligent content generation. The system enables authenticated users to upload or enter study materials and leverage the **Google Gemini AI API** to generate educational resources automatically.

The backend processes incoming requests through secure RESTful APIs and performs the following operations:

- Authenticates users using JWT-based security.
- Stores user information and study resources in MongoDB.
- Sends study content to the Google Gemini AI model through dedicated AI service modules.
- Generates concise summaries for faster revision.
- Creates flashcards for active recall learning.
- Produces multiple-choice quizzes for self-assessment.
- Generates personalized study plans based on user preferences and examination timelines.
- Stores generated learning resources for future retrieval and continuous learning.

By separating authentication, business logic, database operations, and AI processing into modular components, the backend ensures maintainability, scalability, and efficient request handling.

Usage

Student Usage

Students register and log in to the application before accessing AI-powered learning features. After authentication, they can upload study material or provide textual content and choose the desired AI functionality, such as generating summaries, quizzes, flashcards, or personalized study plans. The generated content can be reviewed immediately and stored for future reference.

Administrator Usage

Administrators manage registered users, monitor application activity, oversee system usage, and maintain platform integrity. Administrative privileges include managing user accounts, monitoring AI service utilization, reviewing system logs, and ensuring smooth backend operations.

Outcome

By integrating artificial intelligence with a secure backend architecture, AI StudyBuddy significantly improves the learning experience for students. Manual preparation time is reduced through automated content generation, allowing students to focus more on understanding concepts rather than organizing study materials.

The implementation of JWT authentication, MongoDB data management, and Google Gemini AI integration ensures that educational resources are generated securely, efficiently, and consistently. The modular backend architecture further supports future enhancements, enabling additional AI-powered learning capabilities to be integrated with minimal architectural changes.

As a result, AI StudyBuddy serves as a scalable and intelligent educational platform that enhances productivity, promotes effective revision, and supports personalized learning for students preparing for academic examinations and competitive assessments.

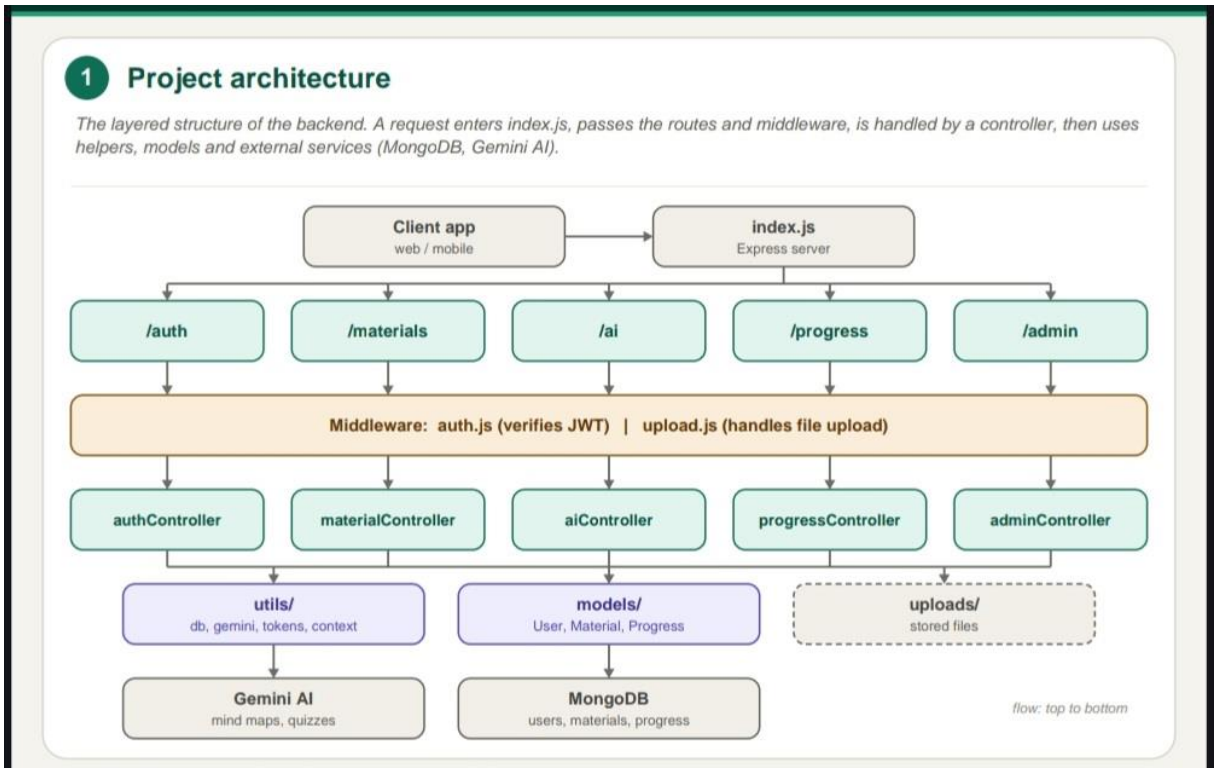
Software Requirements

- **Operating System:** Windows 10/11, macOS, or Linux (supports cross-platform operations).
- **Node.js (v16 or above):** Runtime ecosystem running server logic and routing infrastructure.
- **npm (v8 or above):** Node package manager to control operational dependencies.
- **Express.js:** Lightweight routing web framework to construct backend RESTful entry points.
- **MongoDB:** Document NoSQL engine storing users, posts, categories, comments, and analytics metrics.
- **Postman:** API verification toolkit to assert schema validation responses across administrative route shields.
- **Code Editor:** Visual Studio Code or similar IDE environment.

Hardware Requirements

- **Processor:** Intel Core i5 (8th Gen or above) / AMD Ryzen 5 or equivalent.
- **RAM:** 8 GB minimum (16 GB recommended for concurrent instances of MongoDB, processing layers, and execution nodes).
- **Storage:** 1 GB of available disk workspace.

Project Architecture



Project Architecture Description

- The Study Buddy project follows a three-layer architecture consisting of the Frontend, Backend, and Database.

Frontend Layer

- The frontend provides the user interface through which students interact with the application. It contains pages such as Login, Dashboard, Learning Map, Notes, Quiz, and Analytics. HTML, CSS, and JavaScript are used to create the interface and handle user interactions.

Backend Layer

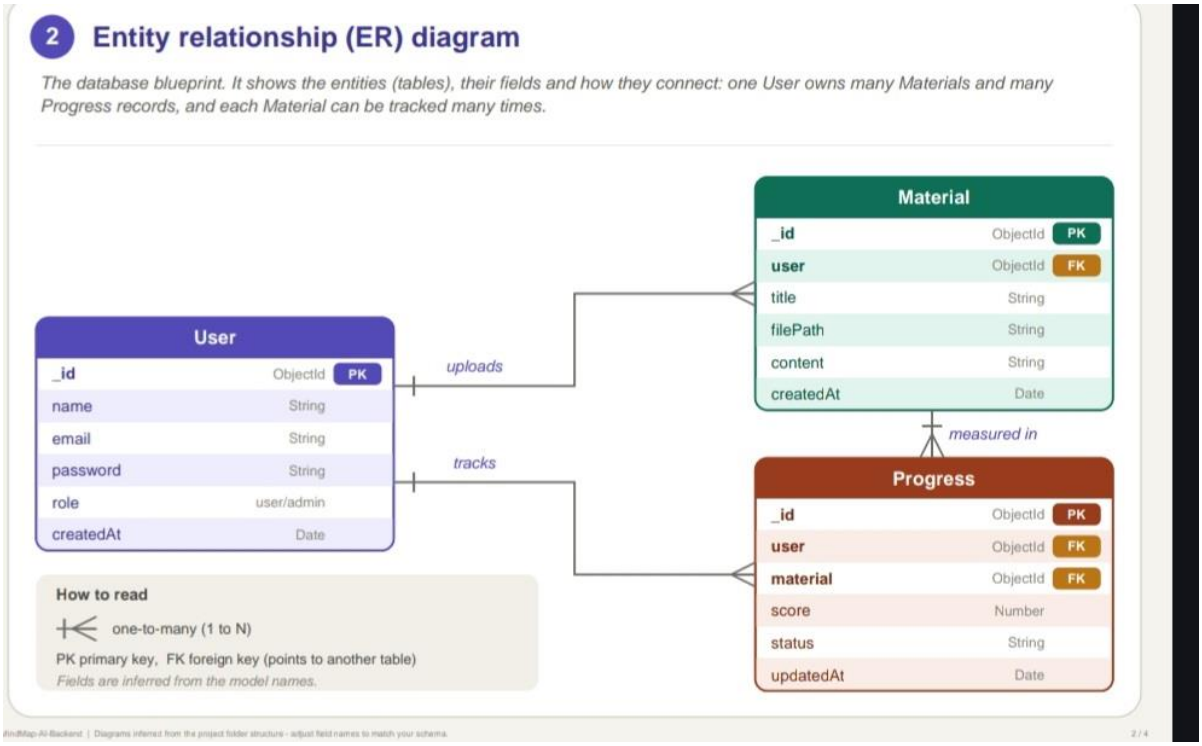
- The backend manages the application's business logic and API requests. Node.js and Express.js can be used to handle authentication, learning activities, notes, and quiz operations. The backend acts as the connection between the frontend and database.

Database Layer

- The database stores important application information such as user details, notes, learning information, quiz data, and progress. MongoDB can be used to store and retrieve this information.

ENTITY-RELATIONSHIP (ER) DIAGRAM DESCRIPTION

1. User Entity



Description:
The User entity stores the details of students and administrators who use the MindMap AI system. It manages user information, login details, roles, and account creation details.
Main attributes: _id, name, email, password, role, createdAt

2. **Summary Entity**

Description:
The Summary entity stores the AI-generated summaries created from the study materials uploaded by users. It helps students quickly understand and revise important concepts.
Main attributes: _id, userId, materialId, summary, createdAt

3. **Study Material Entity**

Description:
The Study Material entity stores learning content uploaded by users. It contains information such as the title, subject, and content that is processed by the AI to generate summaries, flashcards, and quizzes.
Main attributes: _id, userId, title, subject, content, createdAt

4. **Flashcard Entity**

Description:
The Flashcard entity stores AI-generated question-and-answer flashcards based on the uploaded study materials. It helps students with quick revision and active learning.
Main attributes: _id, userId, materialId, question, answer, createdAt

5. **Study Plan Entity**

Description:
The Study Plan entity stores personalized study plans generated by AI. It helps students organize their learning activities according to their study requirements and examination date.

Main attributes: _id, userId, studyPlan, examDate, createdAt

Key Relationships

- **User → Study Material (One-to-Many):** A user can upload multiple study materials.
- **Study Material → Summary (One-to-One):** Each uploaded material can have one AI-generated summary.
- **Study Material → Flashcards (One-to-Many):** One study material can generate multiple flashcards.
- **Study Material → Quiz (One-to-One):** One study material can generate one quiz containing multiple questions.
- **User → Study Plan (One-to-Many):** A user can generate multiple study plans based on different subjects or examination schedules.
- **User → AI Resources (One-to-Many):** All AI-generated summaries, flashcards, quizzes, and study plans are associated with the user who generated them.

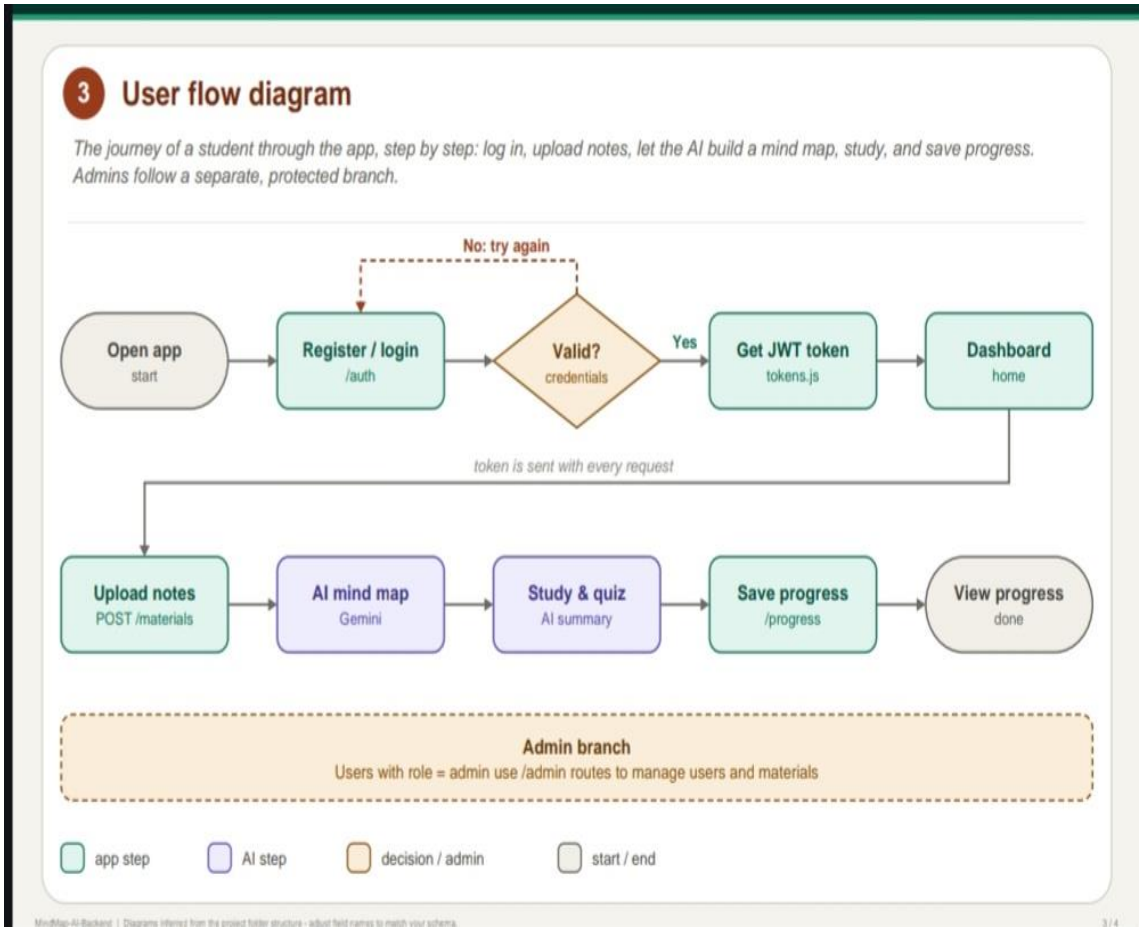
Key Features

- User Management – Stores student and administrator details, including name, email, password, and role.
- Study Material Management – Stores study materials uploaded by students for AI processing.
- AI Summary Generation – Generates and stores summaries from study materials.
- AI Flashcard Generation – Creates questions and answers as flashcards from study materials.
- AI Quiz Generation – Generates quizzes based on uploaded study materials.
- Personalized Study Plans – Creates study plans based on the user's requirements and exam date.
- Data Tracking – Maintains creation dates and user/material relationships for organized data management.

Roles and Responsibilities

The AI Study Buddy system consists of different roles that work together to provide an effective and personalized learning experience. The Student is the main user who registers, logs in, accesses the dashboard, studies topics, creates notes, asks questions, attends AI-generated quizzes, and monitors learning progress. The AI System acts as a virtual study assistant by explaining concepts, generating questions, summarizing notes, creating study plans, and providing personalized recommendations. The Backend manages user requests, authentication, APIs, application logic, and communication between the frontend, AI service, and database. The Database is responsible for storing and retrieving user details, notes,

learning topics, quiz results, and progress records. The Admin manages users, monitors the application, maintains learning content, and ensures the smooth operation of the system. Together, these roles make AI Study Buddy an interactive, organized, and personalized learning platform



User Flow

User Flow – Definition

User Flow is a visual representation of the step-by-step journey followed by a user while using an application. It explains how the user enters the system, navigates between different screens, interacts with features, and completes a particular task.

In the AI Study Buddy project, the user flow describes the complete learning journey of a student. It starts with Registration/Login and moves to the Dashboard, where the student can select a topic, use the AI Assistant, create or read Notes, attend Quizzes, and check Progress and Analytics.

Main User Flow

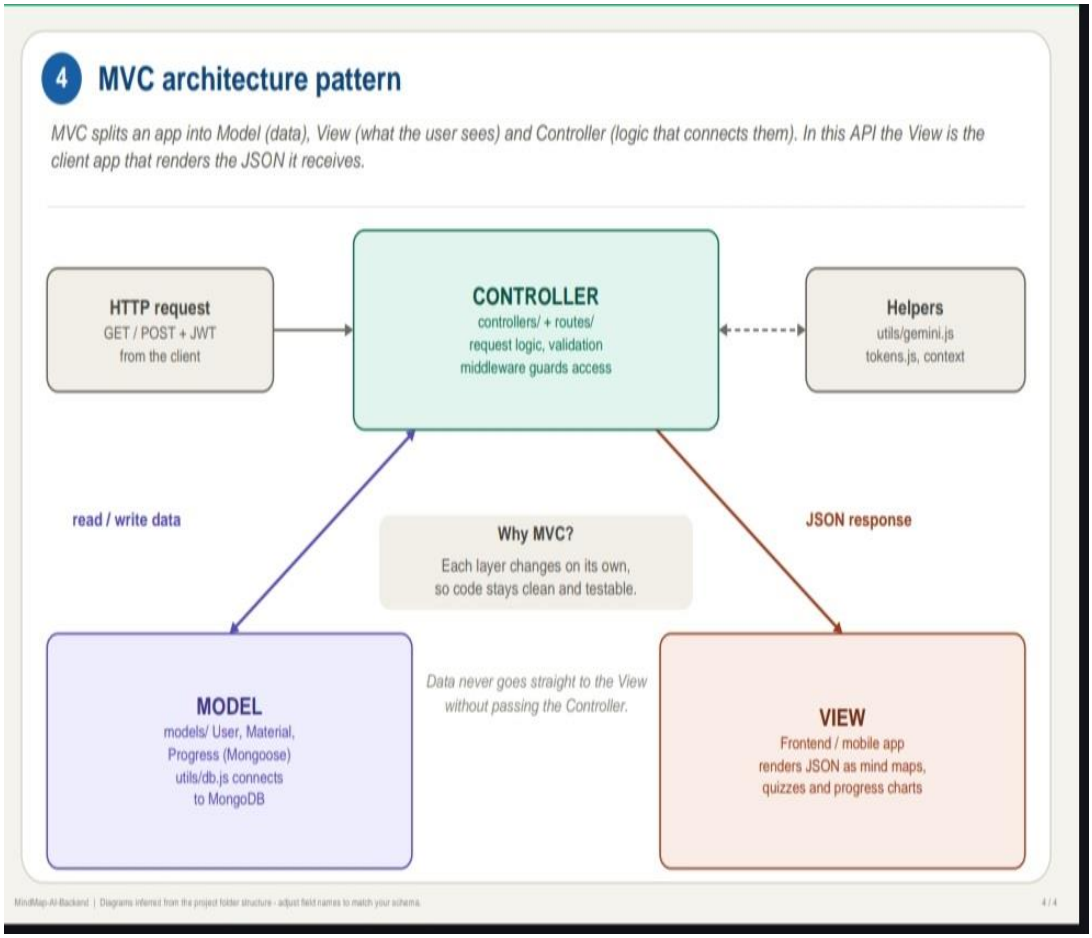
Register/Login → Dashboard → Select Subject → Learning Map → AI Assistant → Notes → Quiz → Results → Progress → Analytics

Importance

1. Shows the complete user journey.
2. Makes navigation easy to understand.
3. Connects all major application features.
4. Helps identify unnecessary steps.
5. Improves the overall user experience.
6. Helps developers design the application structure.
7. Makes the project easier to explain using a flowchart.

MVC Architecture Pattern

MVC Architecture Definition



MVC (Model–View–Controller) Architecture is a software design pattern that divides an application into three main components: Model, View, and Controller. This architecture helps

organize code, improve maintainability, and separate the user interface from business logic and data handling.

Components of MVC:

Model:

The Model manages the application's data, database operations, and business logic. In the Study Buddy project, entities such as User, Study Material, Summary, Flashcard, Quiz, and Study Plan act as Models.

View:

The View represents the user interface that displays information to users. Pages such as Login, Dashboard, Notes, Quiz, and Study Plan are part of the View layer.

Controller:

The Controller acts as an intermediary between the Model and the View. It receives user requests, processes data using the Model, and sends the results to the View.

MVC Workflow

User → View → Controller → Model → Database → Controller → View → User

Advantages of MVC

- Separation of concerns
- Easy maintenance and scalability
- Better code organization
- Reusability of components
- Supports team development

For your AI Study Buddy project, MVC architecture can be implemented using:

Model: MongoDB Models (User.js, StudyMaterial.js, Quiz.js, etc.)

View: HTML, CSS, JavaScript pages

Controller: Express.js Controllers (authController.js, quizController.js, studyPlanController.js)

Project Setup and Configuration

Creating the Project Folder

1. Create a new folder named **AI StudyBuddy** .
2. Open the project folder in **Visual Studio Code**.

3. Initialize the project using Node.js and install the required dependencies.

Project Initialization

Open the project folder in the VS Code terminal and execute:

```
npm init -y
```

Install the required dependencies:

```
npm install express mongoose bcryptjs jsonwebtoken cors dotenv multer @google/genai
```

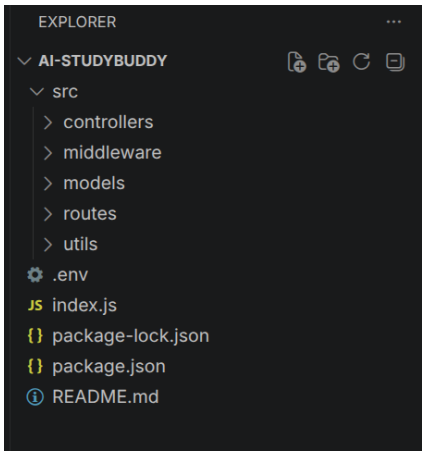
Install Nodemon for development:

```
npm install --save-dev nodemon
```

Create the Following Files

- index.js
- .env
- package.json
- .gitignore

Create the Following Folder Structure



Environment Configuration

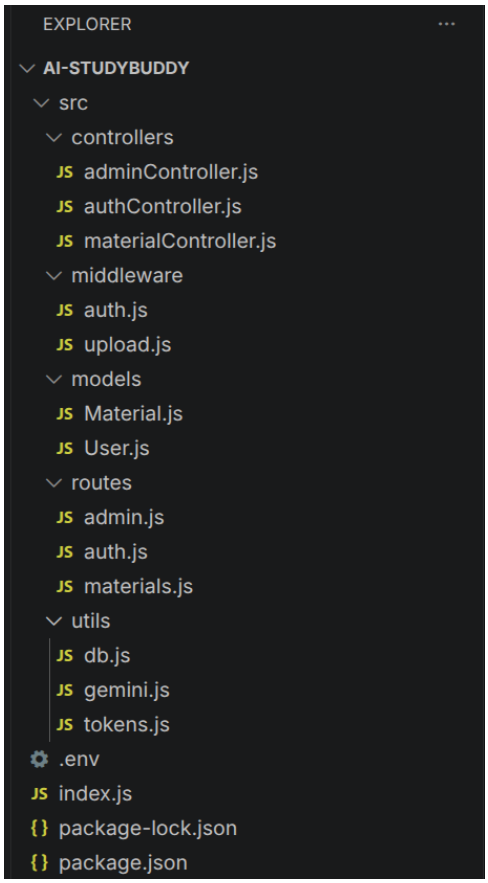
Create a **.env** file in the project root and configure the required environment variables.

```
PORT=5000
MONGO_URI=your_mongodb_connection_string
JWT_SECRET=your_jwt_secret
GEMINI_API_KEY=your_gemini_api_key
```

Backend Development

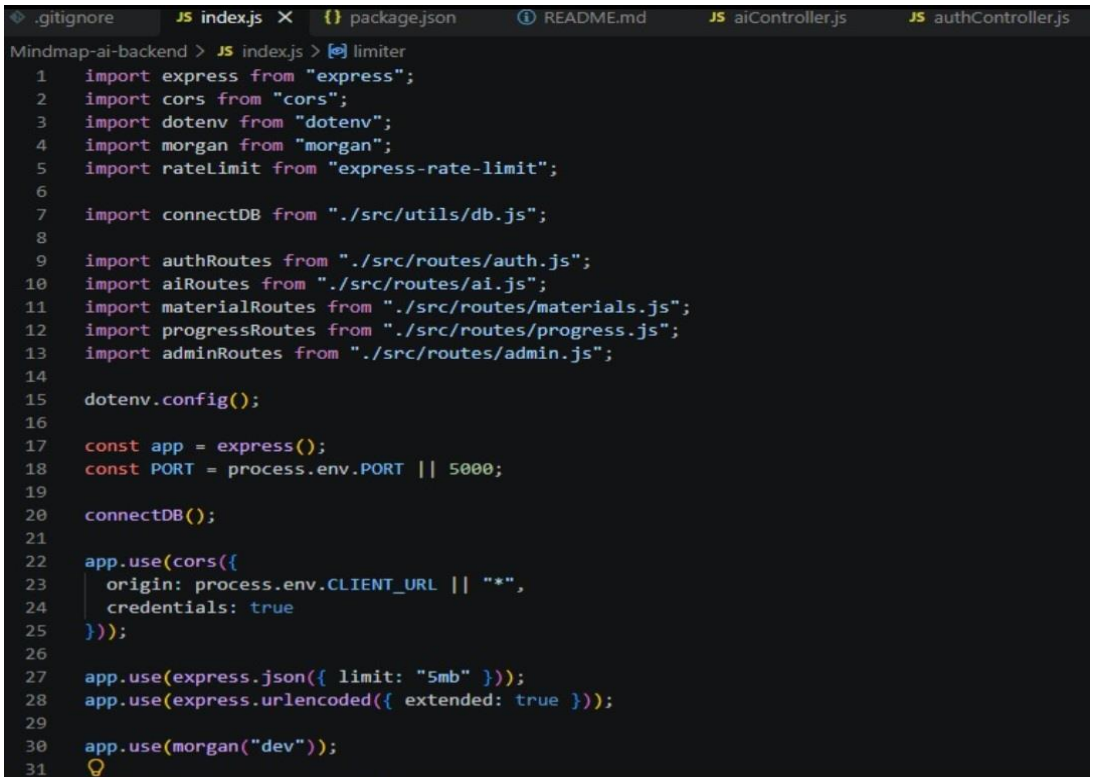
The AI StudyBuddy backend is developed using **Node.js**, **Express.js**, and **MongoDB**, following a modular architecture. The application separates routing, controllers, middleware, models, and utility modules to improve code maintainability and scalability. The backend exposes secure REST APIs for authentication, study material management, AI-powered content generation, and administrative operations.

Backend Structure



Core System Script Implementations

Application Entry Point (index.js)

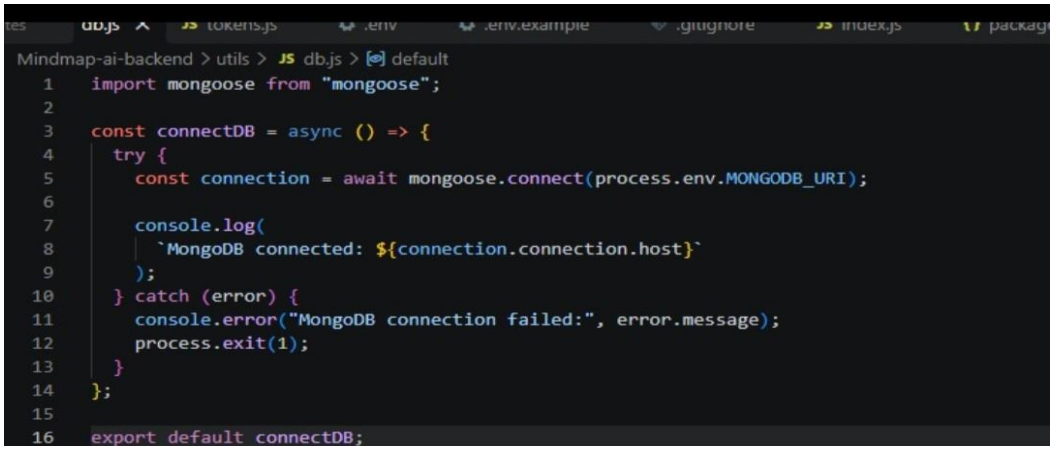


Index.js is the main entry point of the MindMap AI API backend. It initializes the server, loads the required packages, connects the application with the database and API routes, and starts the server for handling user requests.

2. Database Connection (src/utils/db.js)

Db.js – Definition for MindMap-AI Project

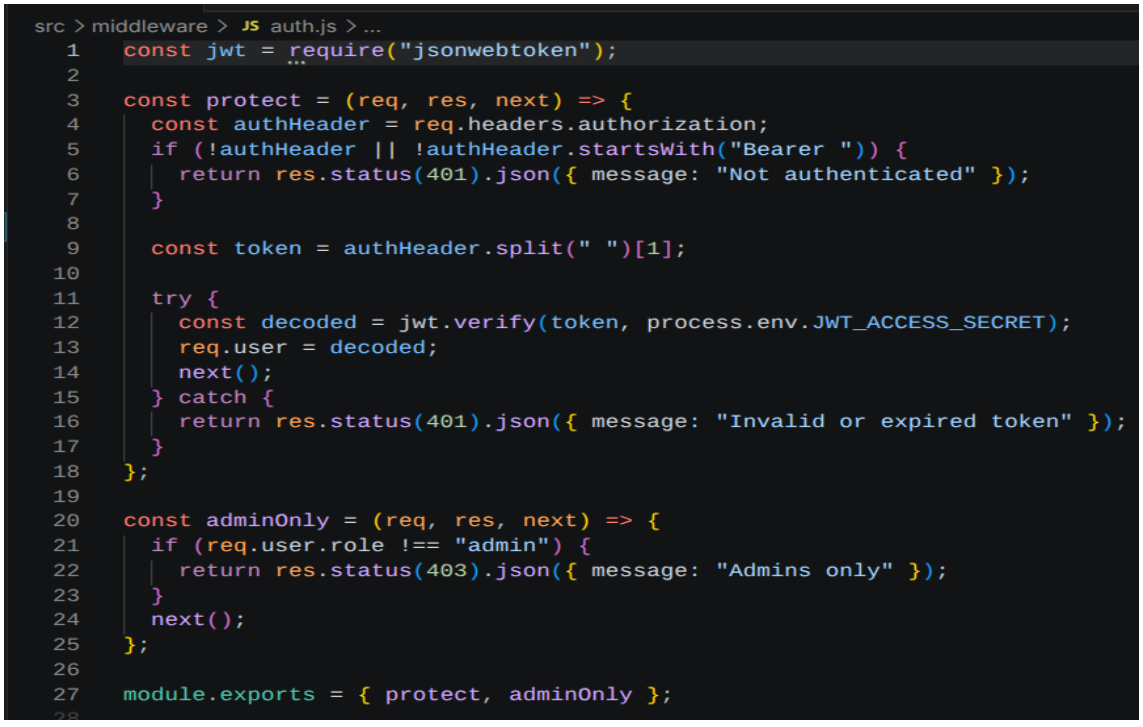
Db.js is the database connection file in the MindMap-AI / AI Study Buddy project. It is responsible for establishing a connection between the backend server and the MongoDB database.



Simple flow:
Backend Server → db.js → MongoDB

3. Authentication Middleware (src/middleware/auth.js)

The authentication middleware verifies JWT access tokens for protected routes. It authenticates users before granting access to secured APIs and includes role-based authorization to restrict administrative operations.



4. AI Integration (src/utils/gemini.js)

This utility integrates the backend with the **Google Gemini 2.5 Flash API**. It sends prompts to the AI model and returns generated responses for summaries, flashcards, quizzes, and study plans.

src > utils > JS gemini.js > ...

```
const { GoogleGenerativeAI } = require("@google/generative-ai");

const genAI = new GoogleGenerativeAI(process.env.GEMINI_API_KEY);
const model = genAI.getGenerativeModel({ model: "gemini-2.5-flash" });

const askGemini = async (prompt) => {
  const result = await model.generateContent(prompt);
  return result.response.text();
};

module.exports = { askGemini };
```

5. File Upload Middleware (src/middleware/upload.js)

The upload middleware handles study material uploads and validates incoming files before they are processed by the by

Database Configuration

The AI StudyBuddy backend uses **MongoDB** as its primary NoSQL database and **Mongoose ODM** for data modeling and schema validation. MongoDB stores user information, uploaded study materials, and AI-generated learning resources in a structured manner. Mongoose simplifies database operations by providing schemas, validation rules, and efficient query handling.

Configure MongoDB

Install Mongoose using the following command:

```
npm install mongoose
```

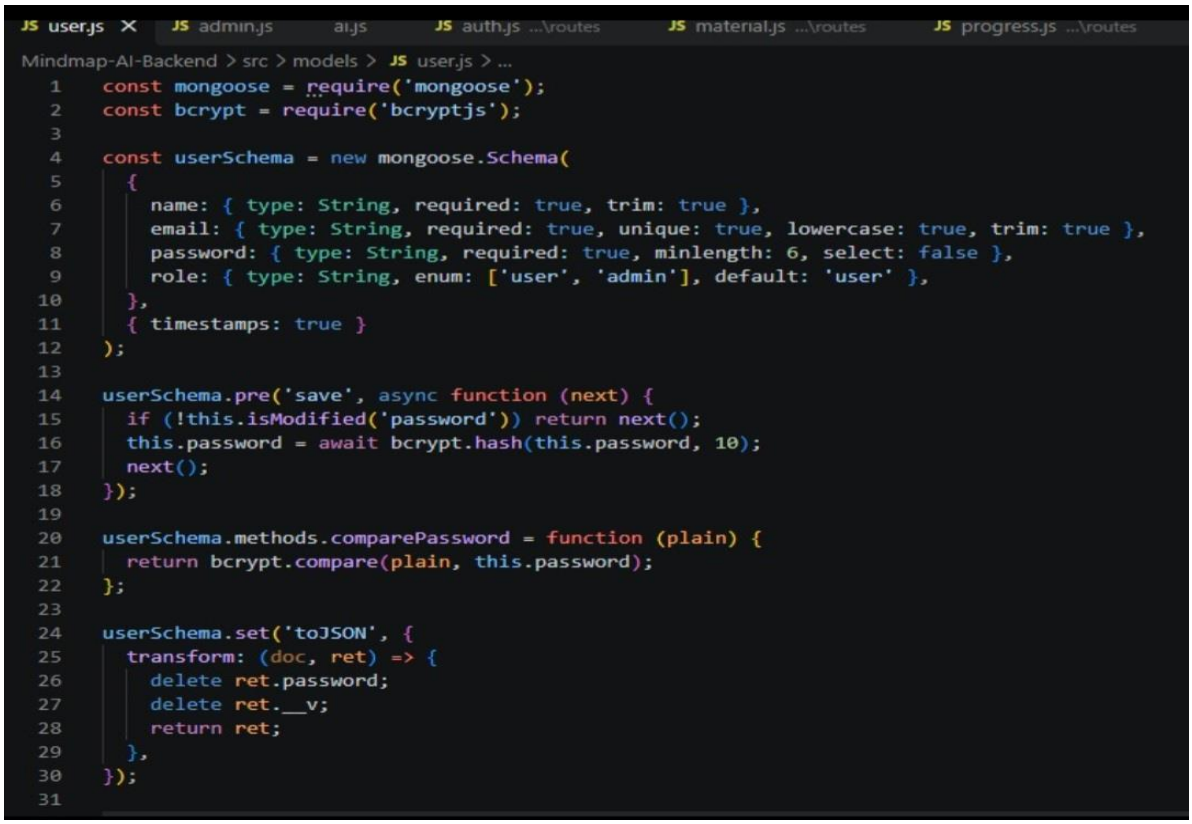
Create Database Connection

Before performing any database operations, establish a connection between the backend and MongoDB using Mongoose. The connection module is initialized during server startup to ensure reliable communication with the database.

Mongoose Schema Definitions

1. User Schema (`models/User.js`)

The **User** schema stores authentication and profile information for registered users. It includes fields such as name, email, password, and role. Passwords are securely encrypted before being stored in the database.



2. Study Material Schema (`models/Material.js`)

The **Study Material** schema stores the learning content uploaded by users. This content acts as the input for AI-powered features such as summaries, quizzes, flashcards, and study plans.

src > models > JS Material.js > ...

```
const mongoose = require("mongoose");

const materialSchema = new mongoose.Schema(
  {
    user: { type: mongoose.Schema.Types.ObjectId, ref: "User", required: true },
    title: { type: String, required: true },
    content: { type: String, required: true }, // raw text content
    filename: { type: String }, // original uploaded file name
    summary: { type: String },
    flashcards: [
      {
        question: String,
        answer: String,
      },
    ],
    quiz: [
      {
        question: String,
        options: [String],
        answer: String,
      },
    ],
    studyPlan: { type: String },
  },
  { timestamps: true }
);

module.exports = mongoose.model("Material", materialSchema);
```

Database Collections

The AI StudyBuddy database consists of the following collections:

- **Users** – Stores user account and authentication details.
- **Study Materials** – Stores uploaded learning materials.

API Route Interface Endpoints

The backend exposes RESTful APIs to interact with the database and AI services.

1. Authentication APIs

Handles user registration, login, JWT authentication, and protected route access.

2. Study Material APIs

Allows users to upload, retrieve, update, and manage study materials.

3. AI Feature APIs

Provides endpoints for:

- AI Summary Generation
- AI Flashcard Generation
- AI Quiz Generation
- AI Study Plan Generation

4. Administrative APIs

Provides administrator-only endpoints for managing users and monitoring system activities.

Project Execution

The AI StudyBuddy backend can be executed locally using Node.js. Before running the application, ensure that MongoDB is running and the required environment variables are configured correctly. The backend exposes REST APIs that can be tested using Postman or Thunder Client.

Running the Backend Server

Install all required dependencies:

```
npm install
```

Start the development server:

```
npm start
```

After successful execution, the server connects to MongoDB and starts listening for incoming API requests.

API Testing

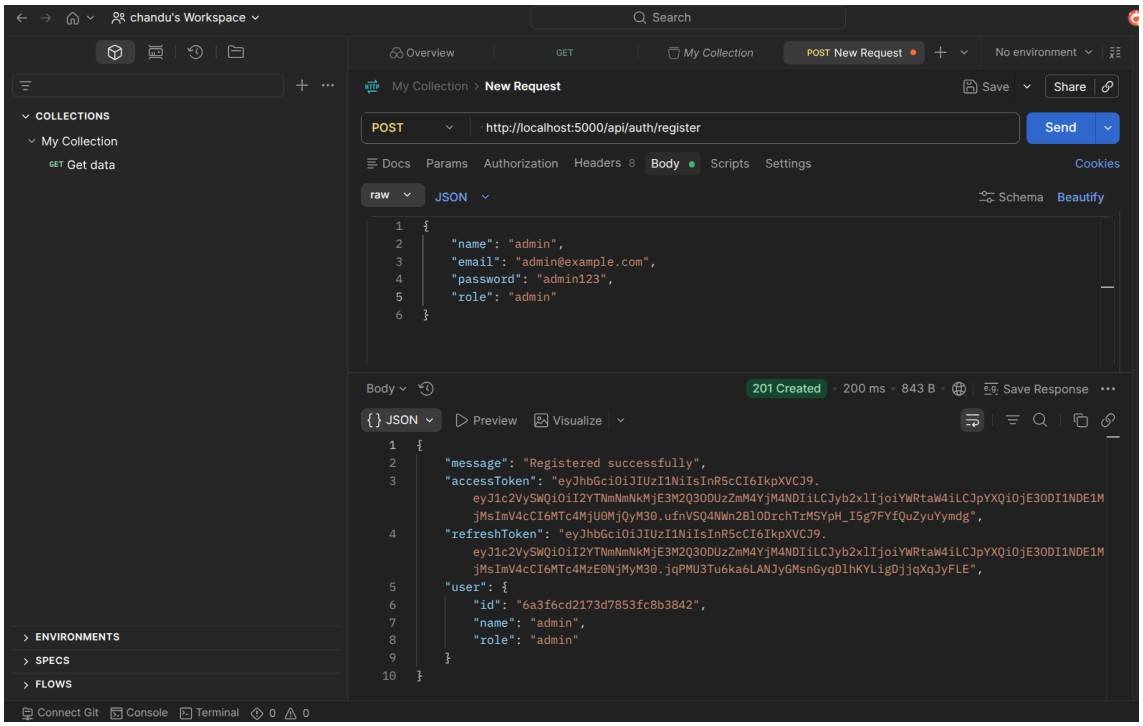
The backend APIs can be tested using **Postman**.

1. User Registration

Registers a new user by accepting the required details and securely storing the encrypted password in the MongoDB database.

Endpoint:

POST /api/auth/register

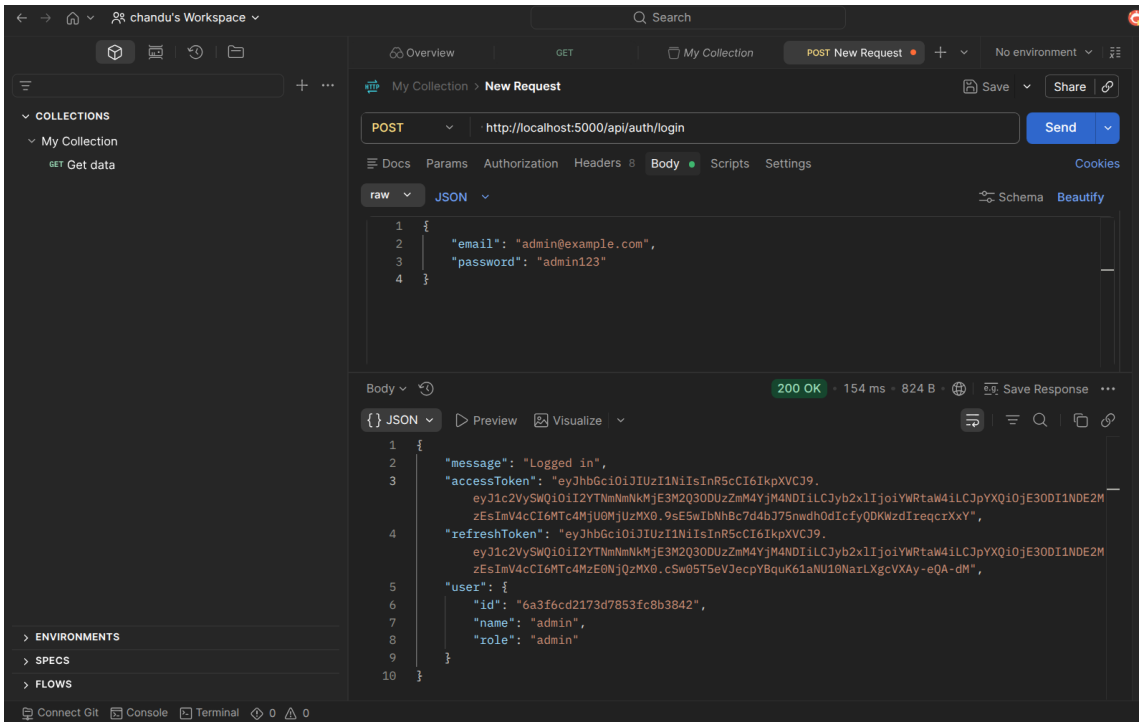


2. User Login

Authenticates the registered user and generates a JWT access token for accessing protected APIs.

Endpoint:

POST /api/auth/login

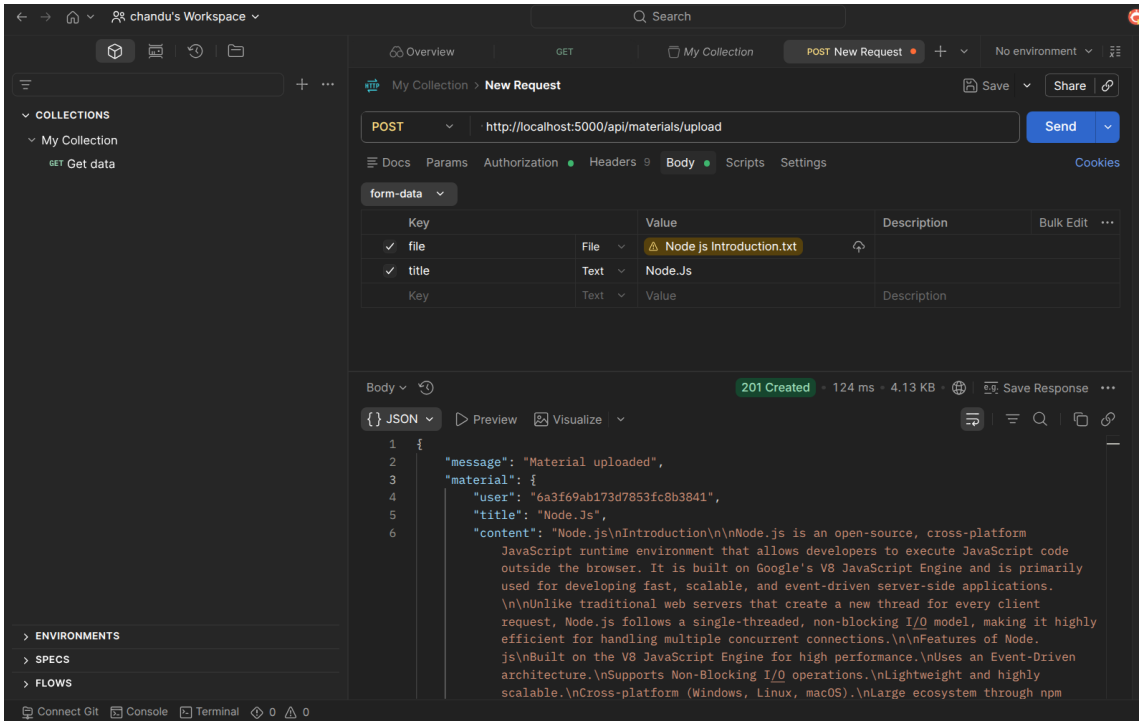


3. Upload Study Material

Allows authenticated users to upload study material that will be used as input for AI-powered content generation.

Endpoint:

POST /api/material/upload

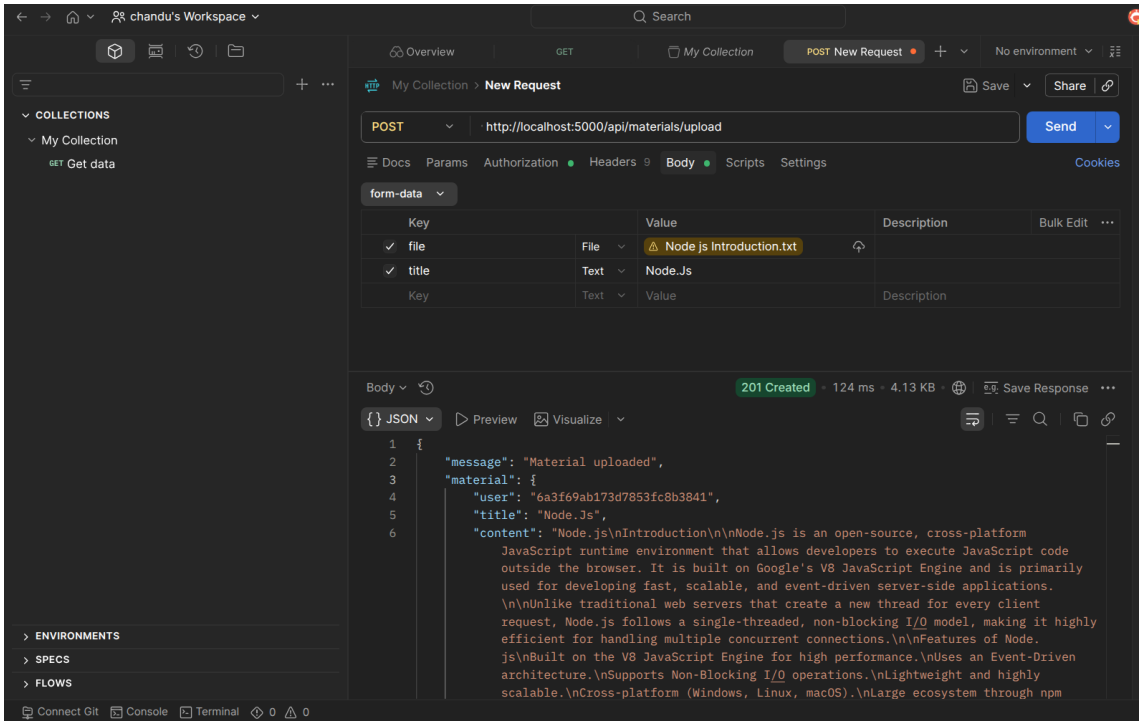


4. Generate AI Summary

Sends the uploaded study material to the Google Gemini API and generates a concise summary.

Endpoint:

POST /api/materials/:id/summarize

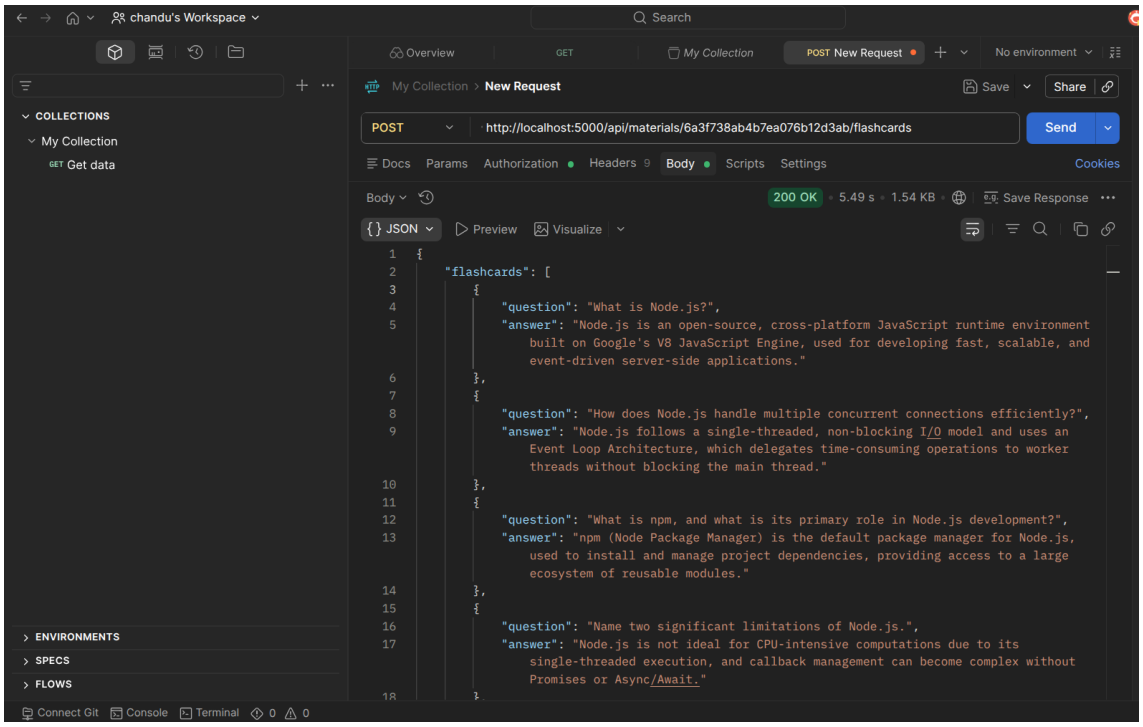


5. Generate AI Flashcards

Creates AI-generated flashcards from the uploaded study material for effective revision.

Endpoint:

POST /api/ai/flashcards

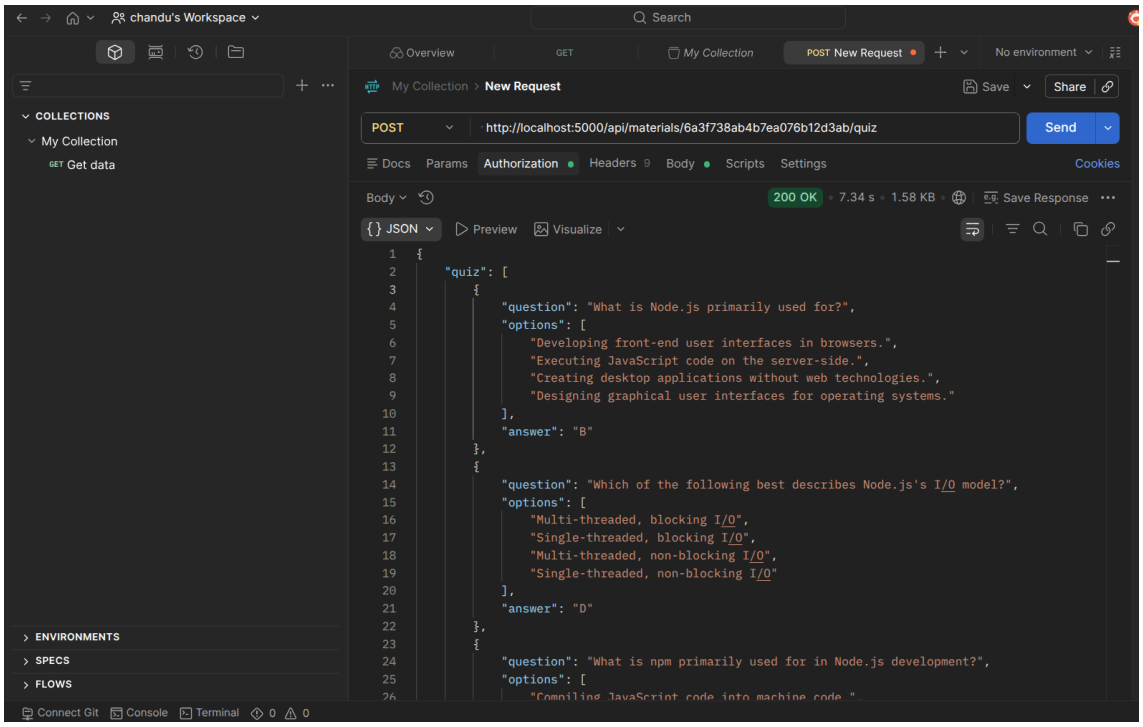


6. Generate AI Quiz

Generates multiple-choice questions from the uploaded study material to help users assess their understanding.

Endpoint:

POST /api/ai/quiz

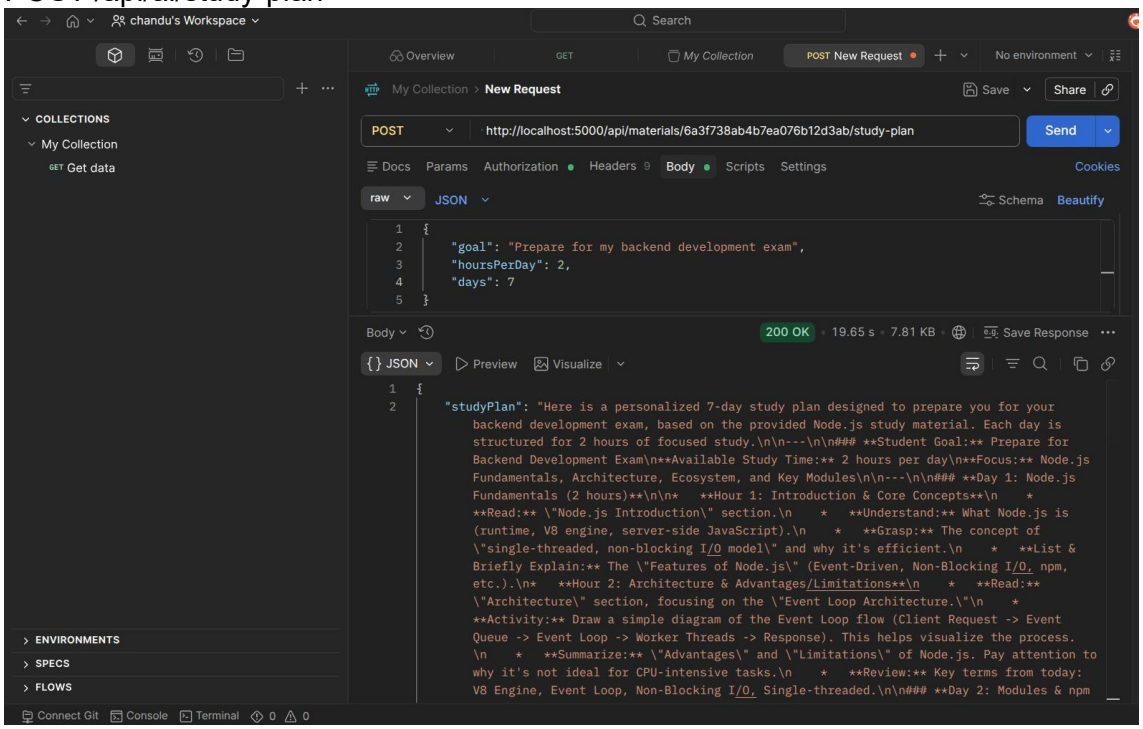


7. Generate AI Study Plan

Creates a personalized study schedule based on the user's learning goals and available study time.

Endpoint:

POST /api/ai/study-plan



API TESTING LINK :

https://docs.google.com/document/d/1v_UKdpGNwdmV-n4vgYWkImiH4mlPPY3l/edit?usp=sharing&ouid=108013194307727689474&rtpof=true&sd=true

DEMO VIDEO LINK :
https://www.mediafire.com/file/wzg2fpxnna4dp3g/Demo+Video.mp4/file

Code link :
https://drive.google.com/drive/folders/1eM_gN4r7VU1kuAC7W3jcfrUHu_2MnL3h?usp=drive_link